Alfiafyvet – Odoo 20

Phase 1 & 2 delivery: DES00134 **alfiafy_product_base** and DES00137 **alfiafy_purchase**

Build and test report · 3 November 2026 · AGM Global Services
Repository folder: `testalfiafyvet/` · Test database: `alfiafy20_test`

2
new modules (DES00134, DES00137)

2
existing modules upgraded from Odoo 19 to Odoo 20

22 / 22
automated tests passed (0 failures, 0 errors)

8
BRD customizations covered (#10–13, 21, 22, 27, 30*)

Module dependency



Dashed boxes are the existing AGM modules, upgraded to Odoo 20 and reused as dependencies. Phase 3–5 modules (stock, sale, POS, partner sourcing, reports, documents) are not part of this delivery.

What was done

- Upgraded `restricted_cost_access` (Hide Cost) and `sale_and_purchase_approvals_steps` from 19.0 to 20.0.
- Built `alfiafy_product_base`: veterinary product master fields, manufacturer and supplier, stock grouping, cost security.
- Built `alfiafy_purchase`: vendor defaults, local / import POs, import tracking, last price and trend, price history popup, sourcing hand-over to receipts and lots, material requisitions, and approval of every PO through the approvals module.
- Created the database `alfiafy20_test`, loaded sample vet data, ran the workflow end to end in the browser (screenshots in this report) and ran 22 automated tests.

* #30 Partner vs Outside sourcing: the purchase → receipt → lot part is delivered here; showing it on POS, sales lines and invoices belongs to DES00139 alfiafy_partner_sourcing.

1. Modules delivered

Module	Ref.	Status	Version	Purpose
alfiafy_product_base	DES00134	New	20.0.1.0.0	Veterinary product master fields, manufacturer / supplier, stock grouping; foundation for all alfiafy_* modules.
restricted_cost_access	used by DES00134	Upgraded 19 → 20	20.0.1.0.0	Hides cost and margin from users outside "Show Cost and Margin" (screens, export, pivots, API).
alfiafy_purchase	DES00137	New	20.0.1.0.0	Purchase type and sourcing, import tracking, price trend and history, sourcing on receipts and lots, material requisitions, PO approval.
sale_and_purchase_approvals_steps	used by DES00137	Upgraded 19 → 20	20.0.1.0.0	Multi-step approval matrix for sale orders, purchase orders and deliveries.

BRD coverage

BRD #	Requirement	Delivered in	How
10	Purchase order approval (all POs)	alfiafy_purchase + approvals	Approval step "Purchase, Amount ≥ 0" installed; any approver at the same sequence may approve.
11	Previous purchase price on PO line	alfiafy_purchase	Last Price + Trend columns; History popup (last 10, all vendors).
12	Import shipment tracking fields	alfiafy_purchase	Import Tracking tab on the PO, repeated on the receipt.
13	Material requisition & approval	alfiafy_purchase	MPR/YYYY/NNNNN: Draft → Waiting Approval → Approved → Done; creates Local RFQs.
21	Additional product master fields	alfiafy_product_base	Arabic name / description, molecule, species, manufacturer, supplier, etc.
22	Product cost visibility restriction	restricted_cost_access	Cost and margin hidden and zeroed for non-authorised users.
27	Local purchase field on PO	alfiafy_purchase	Purchase Type = Local Purchase / Import Shipment.
30	Partner vs outside sourcing	alfiafy_purchase (purchase side)	Vendor → PO → receipt → lot stamped Official Partner / Outside Supplier.

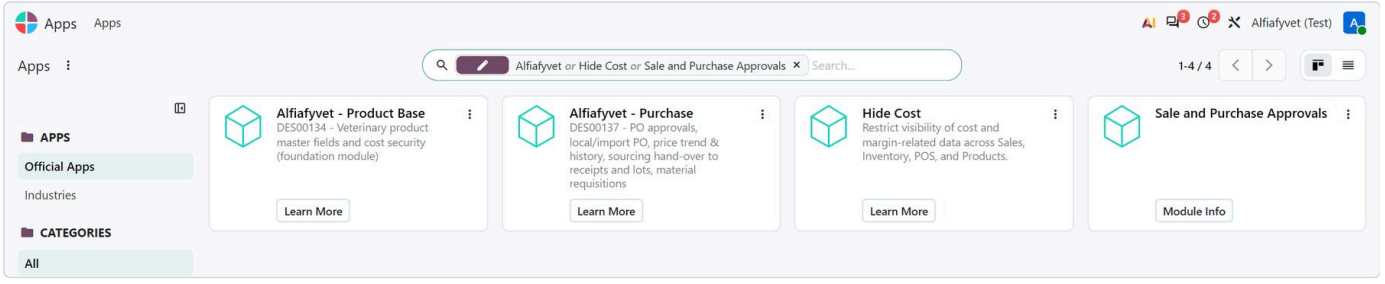
Files

Module	Main files
alfiafy_product_base	models/product_template.py, target_species.py, stock_quant.py · views/product_template_views.xml, stock_quant_views.xml, target_species_views.xml · data/target_species_data.xml · tests/test_product_base.py
alfiafy_purchase	models/res_partner.py, purchase_order.py, purchase_order_line.py, material_requisition.py, stock_picking.py · wizard/purchase_price_history_wizard.py · views/*.xml · data/ir_sequence_data.xml, approval_steps_data.xml · tests/test_alfiafy_purchase.py
restricted_cost_access	models/cost_restriction_mixin.py (new), product.py, sale_order.py, pos_order.py, stock.py · views updated · tests/test_cost_restriction.py (new)
sale_and_purchase_approvals_steps	security/ir.access.csv (replaces ir.model.access.csv + ir.rule) · models/approvals.py, stock_picking.py · views/stock_picking_view.xml · tests/test_approval_steps.py (new)

2. Test environment

Item	Value
Odoo	20.0 community + enterprise, Python 3.12 (`.venv20`), PostgreSQL 16
Database	`alfiafy20_test` on `http://localhost:8021` (the normal Odoo 20 port 8020 was not touched)
Addons path	`enterprise-20.0, odoo/addons, testalfiafyvet`
Company	Alfiafyvet (Test), Qatar, currency QAR, warehouse “Main Warehouse (Doha)”, lots enabled
Sample data	3 vendors (2 Official Partner / import, 1 Outside / local), 5 vet products, 9 purchase orders, 9 lots, 3 material requisitions

Login	Password	Role in the test	Sees cost?	Approves POs?
admin	admin	Management (administrator)	Yes	Yes (step 10)
accountant	Test@1234	Accountant, Purchase / Inventory administrator	Yes	Yes (step 10)
buyer	Test@1234	Purchase Officer (purchase user)	No	No
pharmacist	Test@1234	Branch pharmacist (inventory, sales, POS user)	No	No



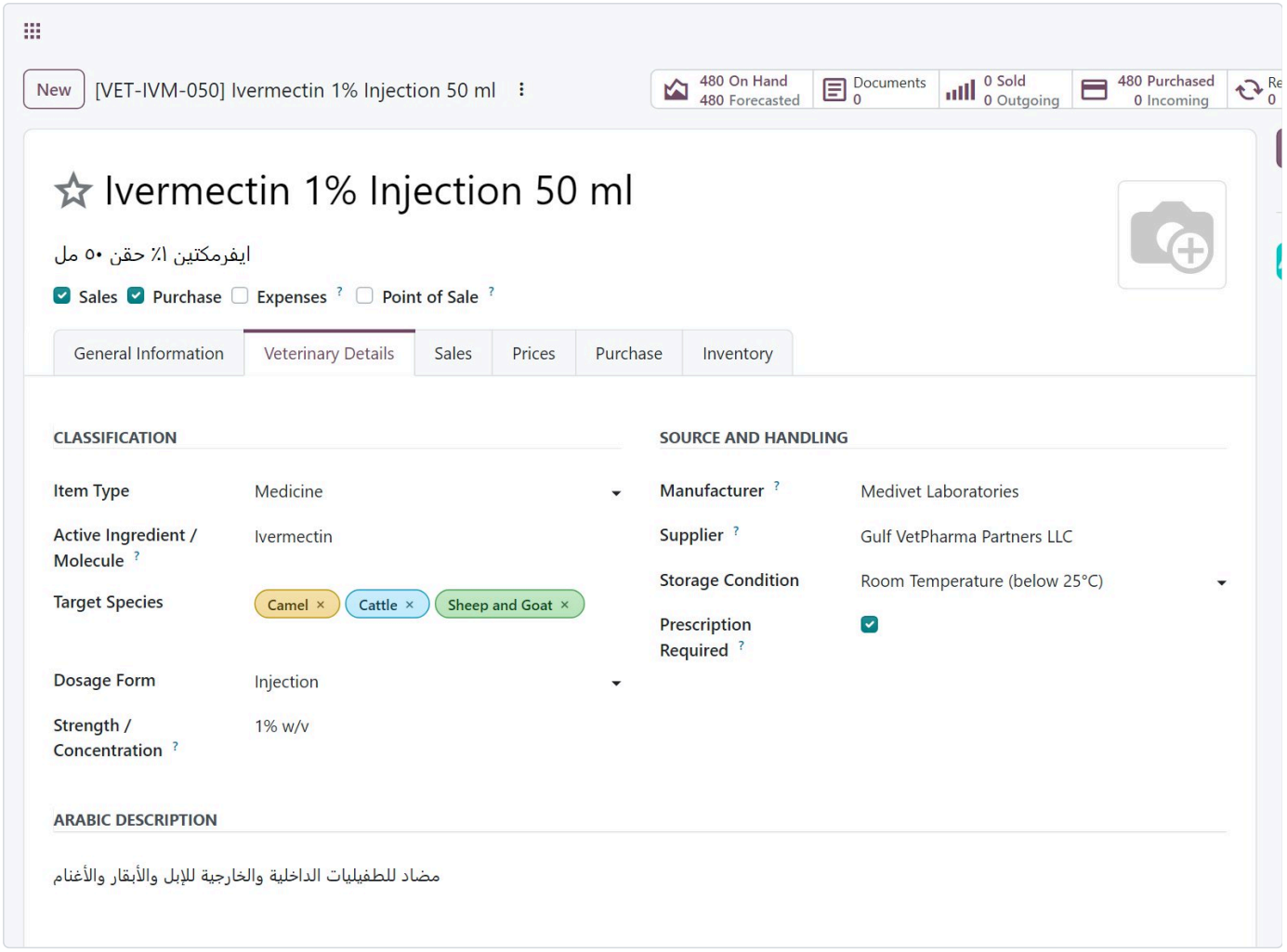
Apps – the four modules installed in alfiafy20_test.

3. DES00134 – alfiafy_product_base

Adds the veterinary product master data on `product.template` (variants inherit it) and makes stock reportable by manufacturer and supplier. Cost security is provided by the upgraded `restricted_cost_access`, which this module depends on.

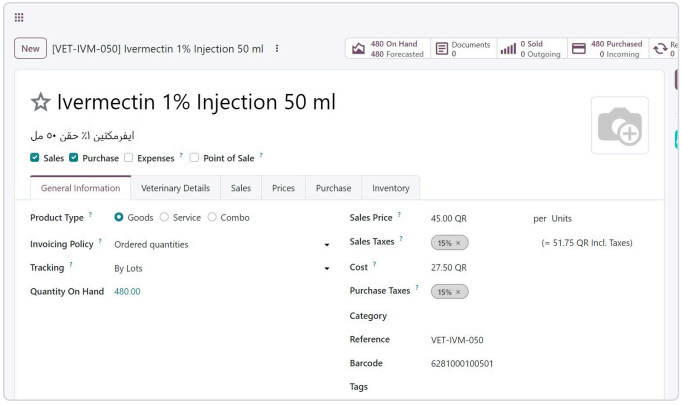
Technical name	Label	Type	Notes
`name_ar`	Arabic Name	Char	Shown under the product name; searchable
`description_ar`	Arabic Description	Text	
`active_ingredient`	Active Ingredient / Molecule	Char	Generic compound name; searchable (trigram index)
`target_species_ids`	Target Species	Many2many	New model `alfiafy.target.species`: Camel, Horse, Cattle, Sheep and Goat, Poultry, Dog, Cat, Pet Birds
`vet_item_type`	Item Type	Selection	Medicine, Vaccine / Biological, Feed Supplement, Equipment, Instrument, Other
`dosage_form`	Dosage Form	Selection	Injection, Tablet / Bolus, Oral Liquid, Powder, Paste, Topical, Drops, Other
`strength`	Strength / Concentration	Char	e.g. 1% w/v
`storage_condition`	Storage Condition	Selection	Room temperature, Cool and dry, Refrigerated 2–8°C, Frozen
`prescription_required`	Prescription Required	Boolean	
`manufacturer_id`	Manufacturer	Many2one res.partner	BRD 1.5 #8: separate from supplier
`supplier_id`	Supplier	Many2one res.partner	Also used as default vendor by material requisitions
`stock.quant.product_manufacturer_id / product_supplier_id`	Manufacturer / Supplier on stock	stored related	Filter and group on-hand stock

> The design table pasted for DES00134 was cut off after `target_species`. The other fields come from BRD 1.5 #8 (Arabic name and description, manufacturer, supplier) plus standard vet-pharmacy data (item type, dosage form, strength, storage, prescription). Please send the complete field list if it differs; adding or removing a field is a small change.

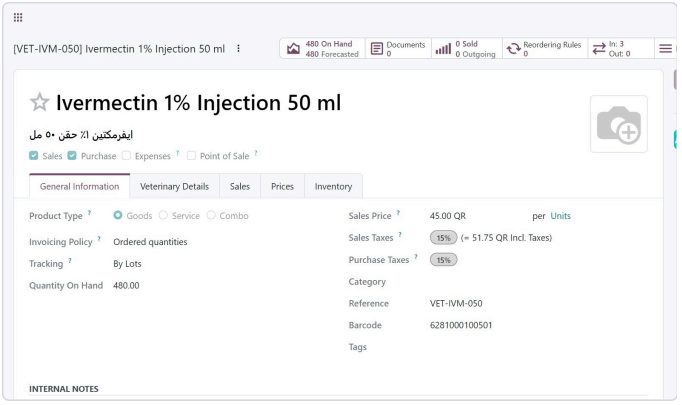


Product form – Veterinary Details tab: Arabic name under the title, classification, target species tags, manufacturer and supplier, Arabic description.

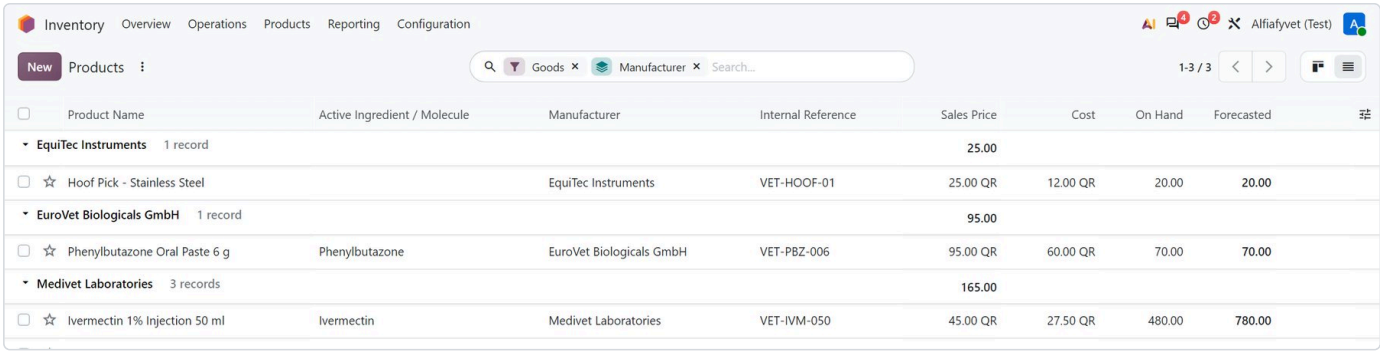
Cost security: management vs branch user



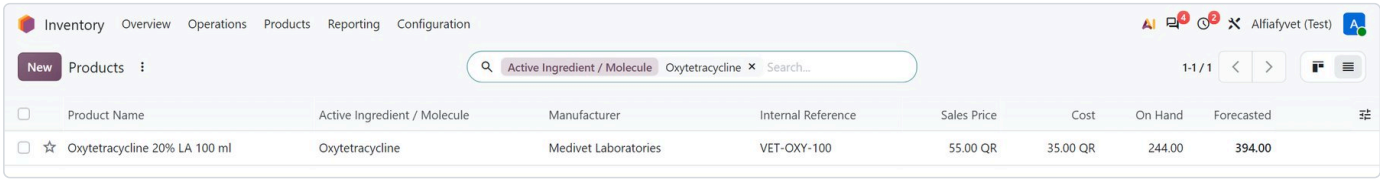
admin (Show Cost and Margin): Cost 27.50 QR and the Purchase tab are visible.



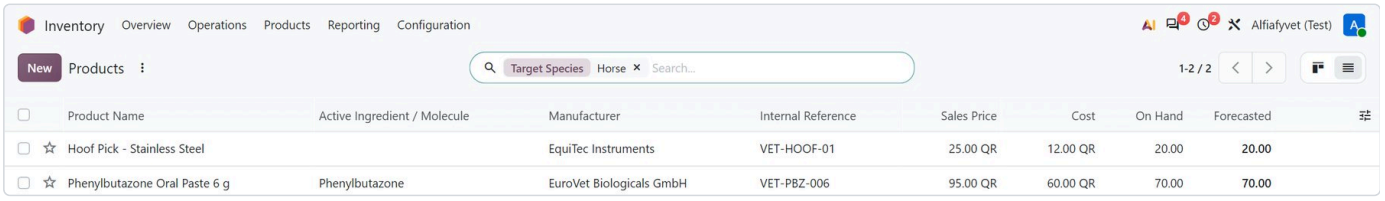
pharmacist: same product, no Cost field, no Purchase tab (vendor prices). Sales price stays visible.



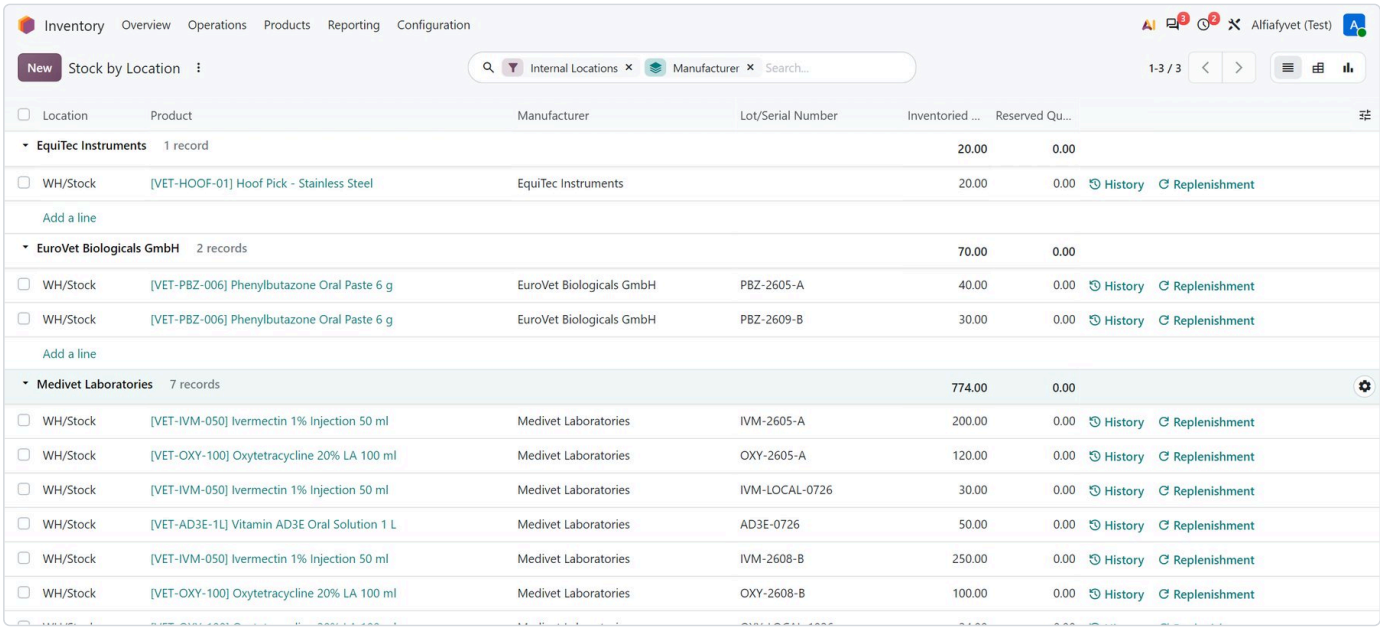
Products list grouped by Manufacturer with the Active Ingredient and Manufacturer columns.



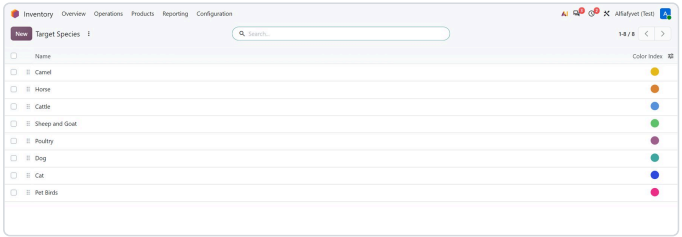
Search by Active Ingredient (“Oxytetracycline”).



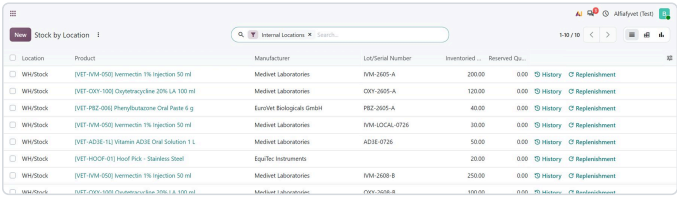
Search by Target Species (“Horse”).



Inventory > Stock grouped by Manufacturer (with lots), as required by BRD 1.5 #8.



Inventory > Configuration > Products > Target Species.



Stock as pharmacist: quantities and lots only, no valuation.

4. restricted_cost_access – upgrade 19.0 → 20.0

Same behaviour and the same group (`restricted_cost_access.group_manager_cost`, "Show Cost and Margin"), adapted to the Odoo 20 API:

Odoo 20 change	What was changed in the module
`read_group()` is `@typing.final` and returns tuples; the web client uses `formatted_read_group()` / `formatted_read_grouping_sets()`	The old `read_group` / `web_read_group` overrides (which would break in 20) were replaced by overrides of the two new methods. Pivot, graph and grouped-list totals are zeroed.
A field used in a view but missing from `fields_get()` crashes the Owl client	Restricted fields are no longer popped; they are flagged not exportable / searchable / sortable / groupable and lose their aggregator, so they still disappear from Export, filters and measures.
`product.product_variant_easy_edit_view` removed	Inheritance removed; the variant form is built on the template form and inherits the restriction.
sale_margin: margin now sits inside the order-line price column; new "Projected Margins" report	Order-line cost / margin columns, quotation-template cost and the Projected Margins list are restricted; the report action is limited to the cost group; `sale.report.purchase_price` is redacted.
— (code quality)	About 900 lines of copy-pasted overrides replaced by one mixin, `restricted.cost.mixin`; each model only declares its sensitive fields.
—	Administrator added to "Show Cost and Margin" on install; 5 automated tests added.

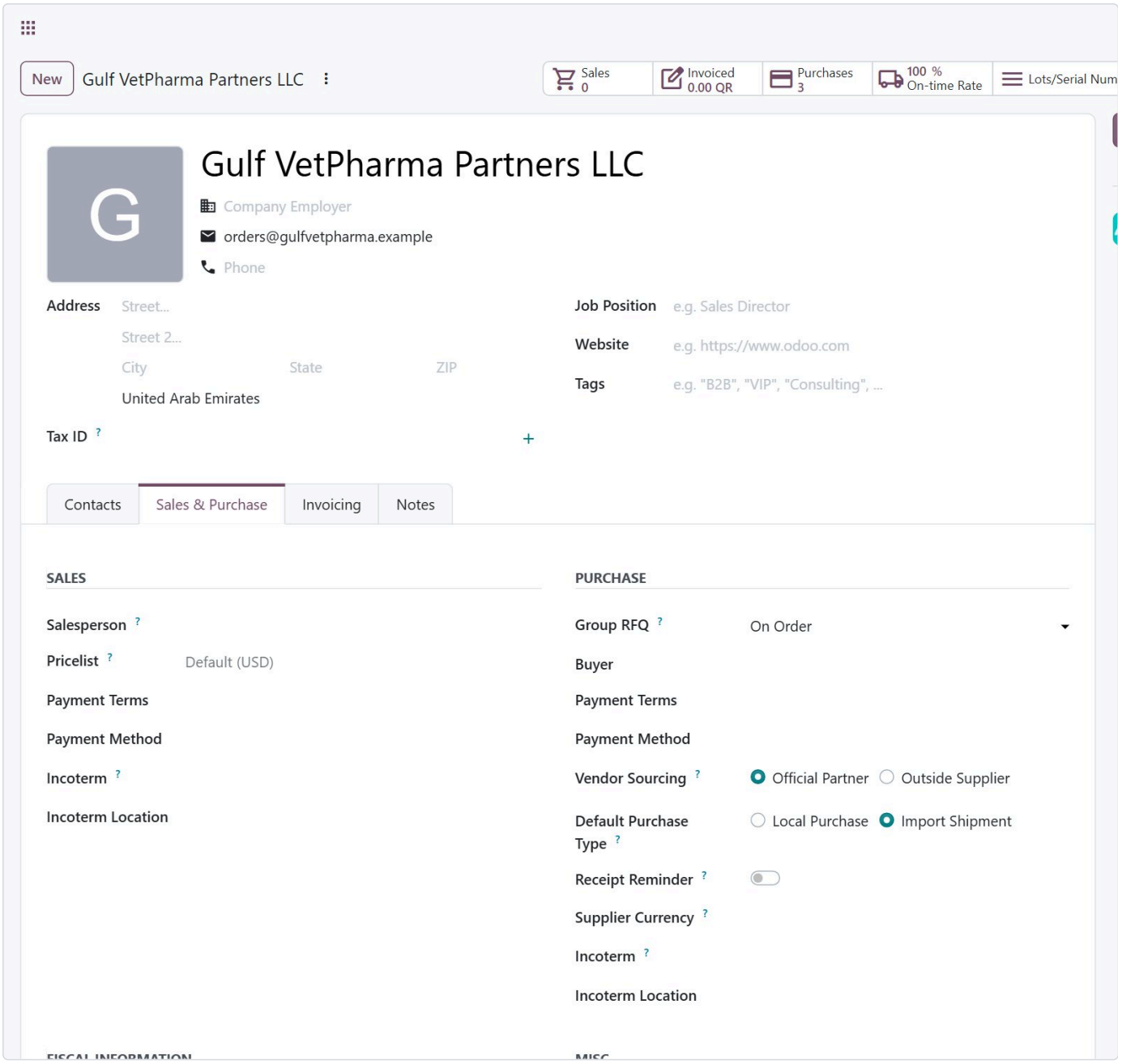
5. sale_and_purchase_approvals_steps – upgrade 19.0 → 20.0

Odoo 20 change	What was changed in the module
`ir.model.access` and `ir.rule` merged into `ir.access`	`security/ir.model.access.csv` and the multi-company record rule replaced by `security/ir.access.csv`.
Alternative POs (`alternative_po_ids`) moved from purchase_requisition to `purchase_alternative`	Dependency switched to `purchase_alternative` (the waiver "fewer than 3 alternatives" check keeps working). Unused `sale_expense` dependency removed (it pulled in HR Expenses).
Picking Validate buttons are driven by the computed `validate_button_style`	The 19.0 xpaths on the two Validate buttons no longer match; replaced by a compute override that hides Validate while a delivery waits for approval.
`_()` outside a request cannot find the language	Module-level helper functions now use `record.env._()` (removed "no translation language detected" warnings with stack traces).
— (bug fix)	Approval step currency now follows a change of the company currency.
—	3 automated tests added (sale threshold, direct confirm under threshold, purchase rejection with reason).

6. DES00137 – alfiafy_purchase

6.1 Vendor defaults

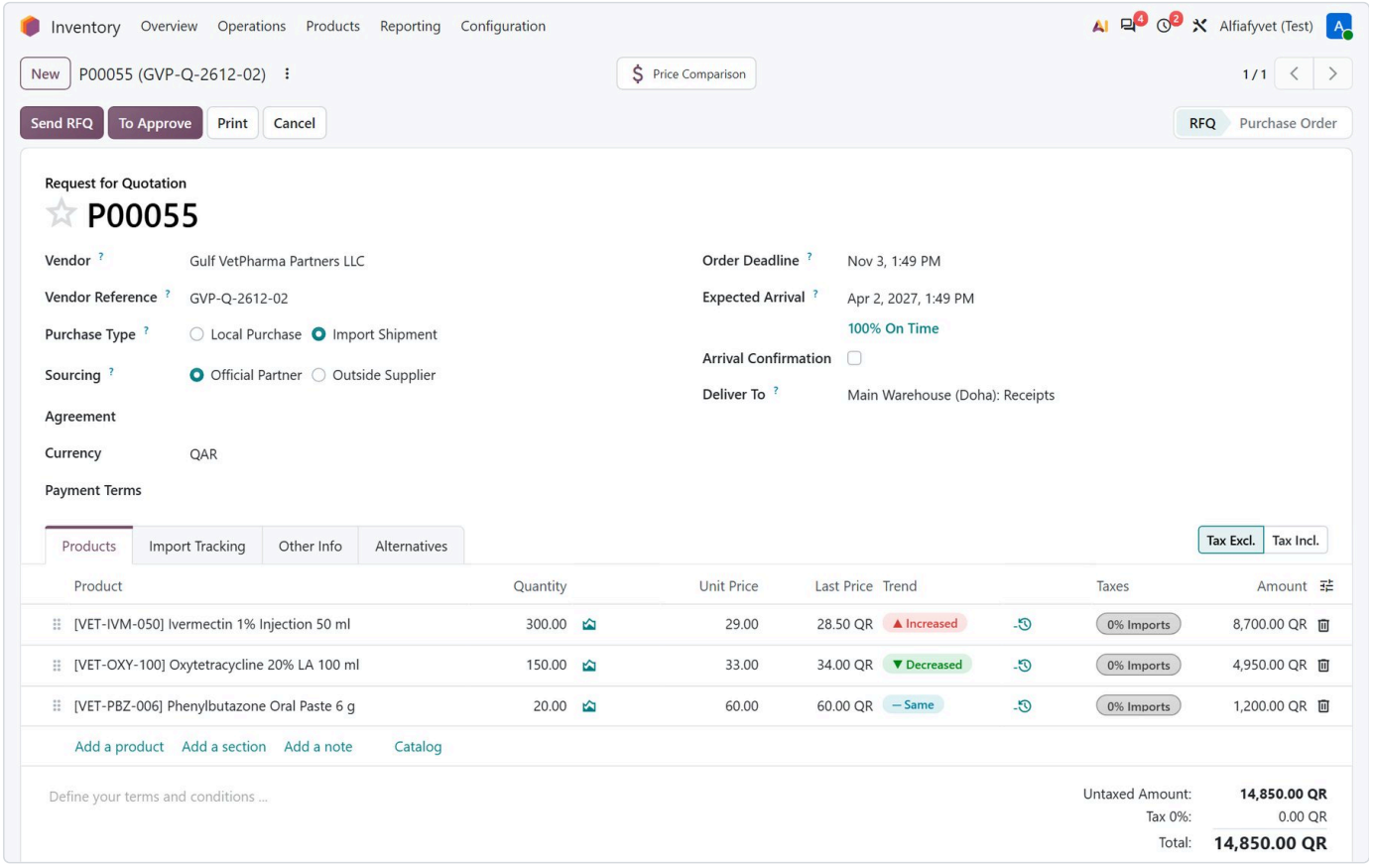
Two fields on the vendor (Sales & Purchase tab). Contacts of the vendor company follow the company.



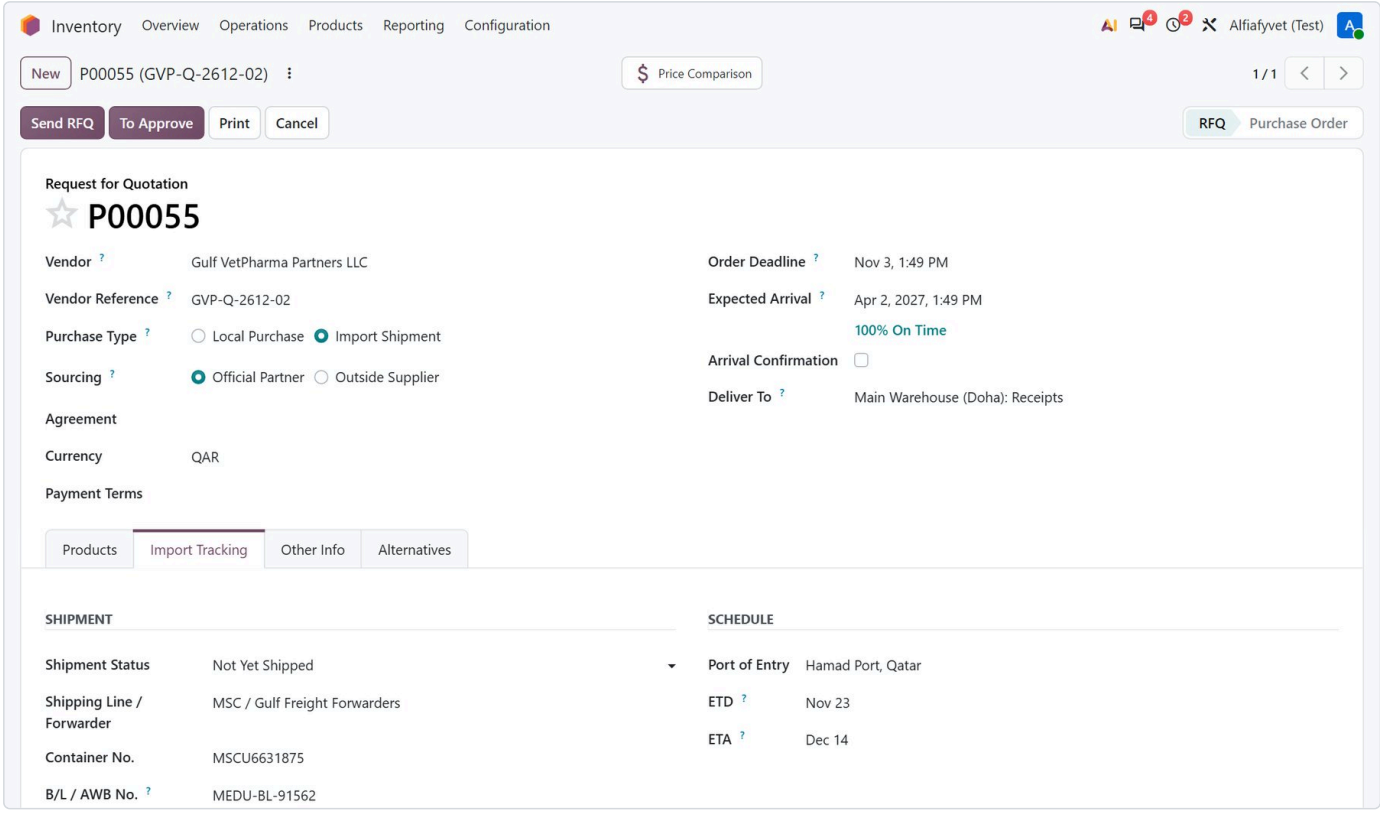
Vendor – Vendor Sourcing = Official Partner, Default Purchase Type = Import Shipment.

6.2 Purchase order: type, sourcing, last price and trend

Purchase Type and Sourcing are pulled from the vendor when it is selected and can be changed. Each line shows the last confirmed price from the same vendor (converted to the line's unit and currency) and the trend.

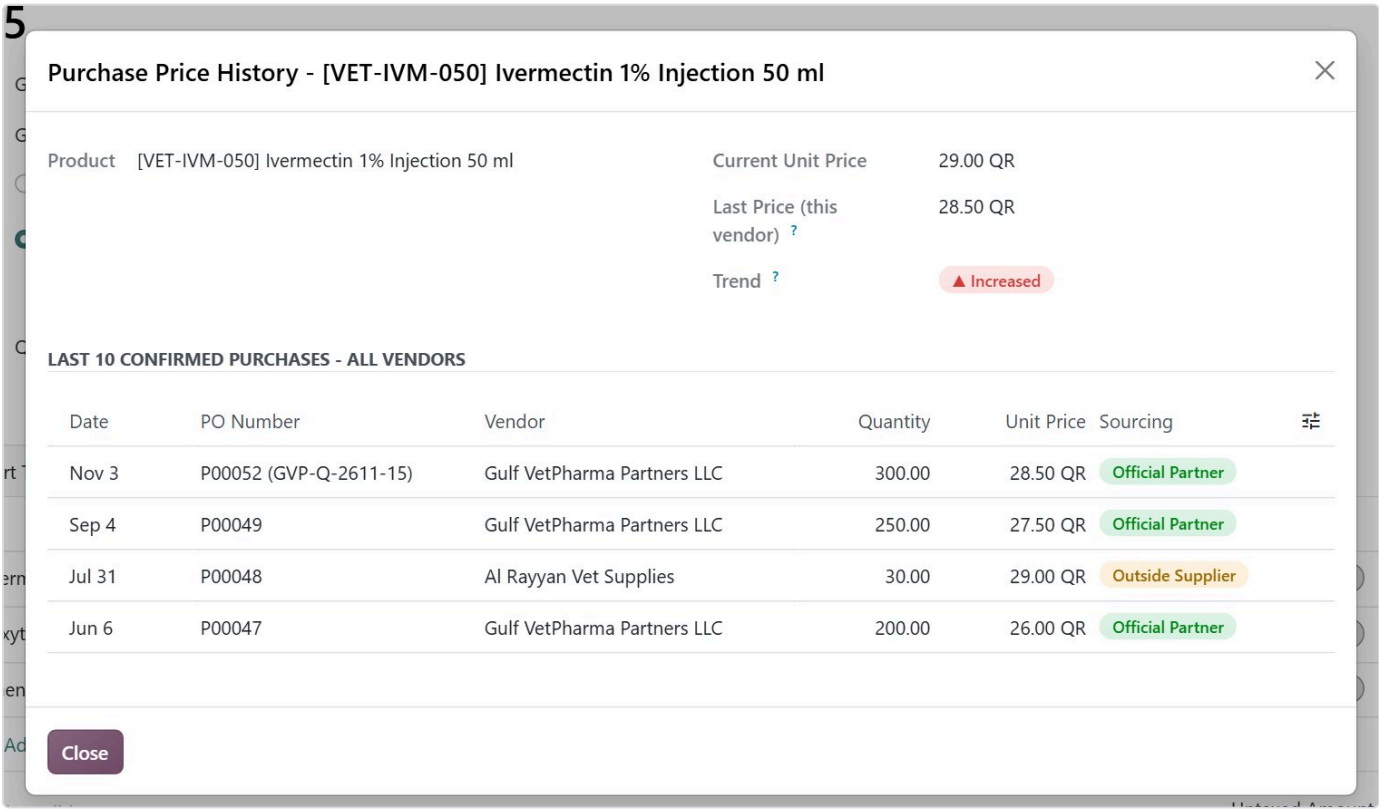


Draft import RFQ P00055: Ivermectin 29.00 vs 28.50 → ▲ **Increased**; Oxytetracycline 33.00 vs 34.00 → ▼ **Decreased**; Phenylbutazone 60.00 vs 60.00 → — **Same**. The clock icon on each line opens the price history.



Import Tracking tab (only shown when Purchase Type = Import Shipment): shipment status, forwarder, container, B/L / AWB, tracking no., port of entry, ETD, ETA.

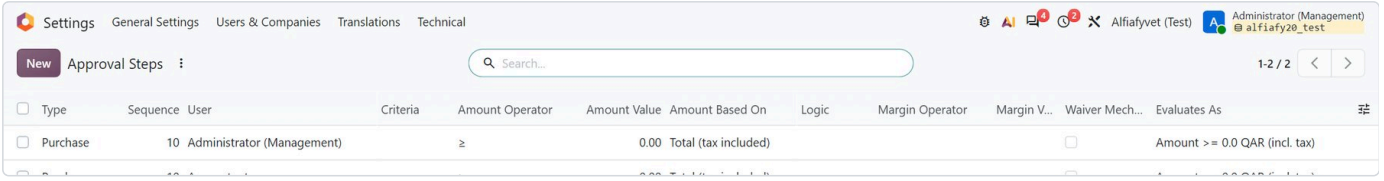
6.3 Price history popup



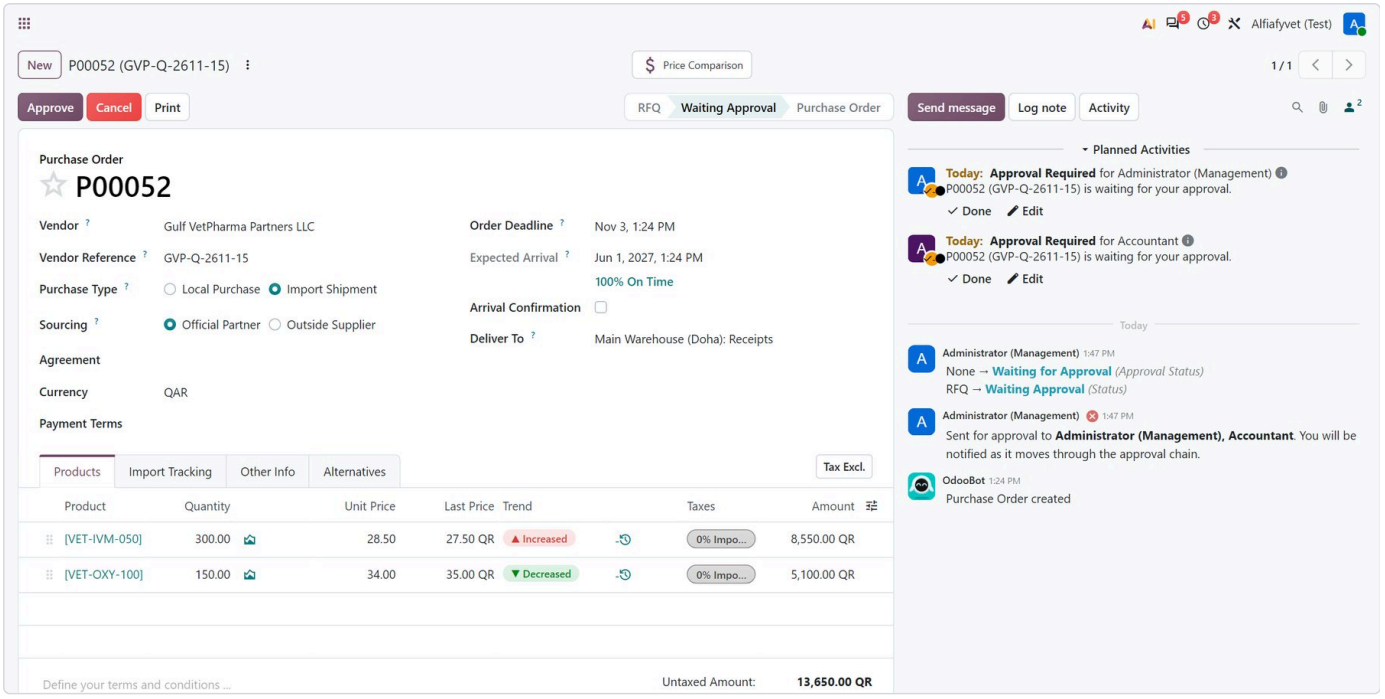
Purchase Price History: last 10 confirmed purchases of the product across all vendors, with date, PO, vendor, quantity, unit price and sourcing; opened from the line without leaving the order.

6.4 Approval of every purchase order

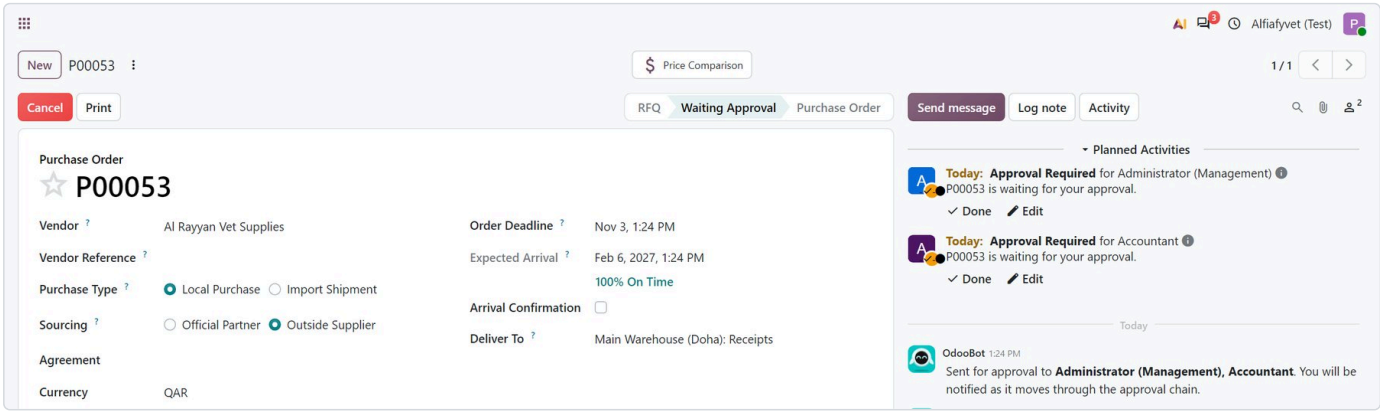
The module installs an approval step *Purchase · Amount ≥ 0*, so "Confirm Order" is replaced by "To Approve" on every RFQ. In the test DB the administrator and the accountant are both at sequence 10, so either one can approve (BRD 1.4 #1).



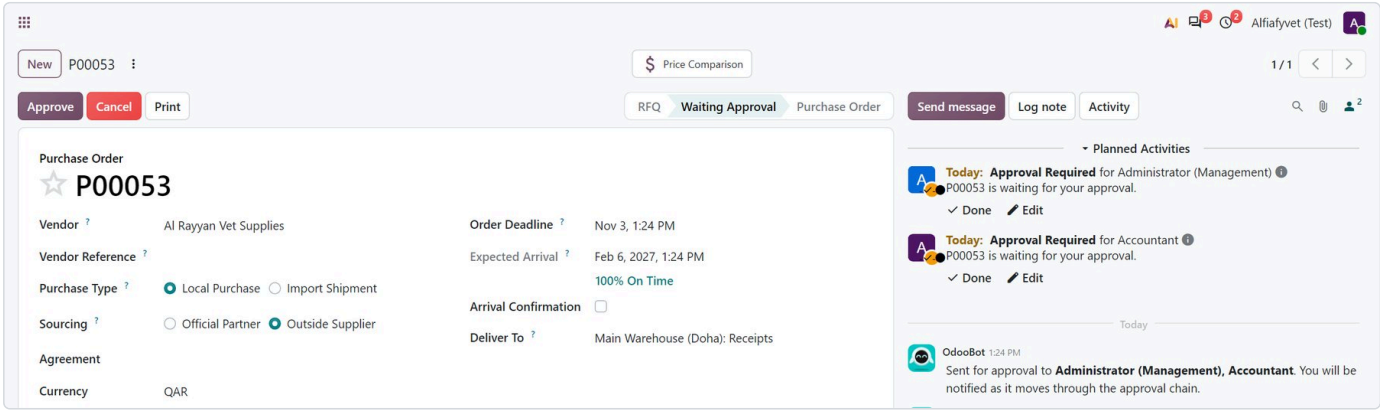
Settings > Technical > Approval Steps – two approvers at the same level.



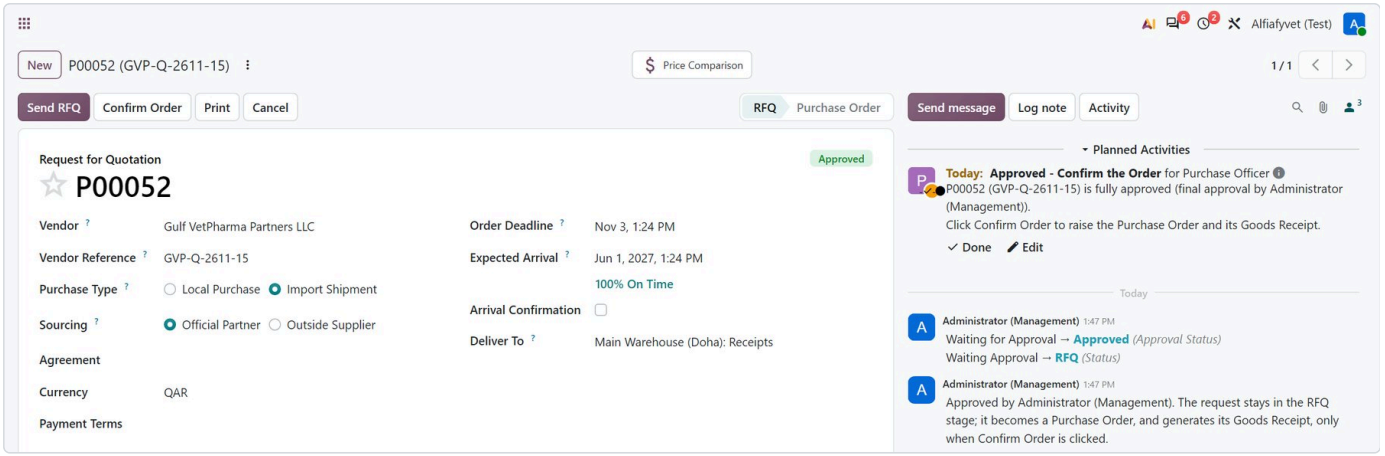
After "To Approve": status Waiting Approval, lines locked, activities created for both approvers, requester notified.



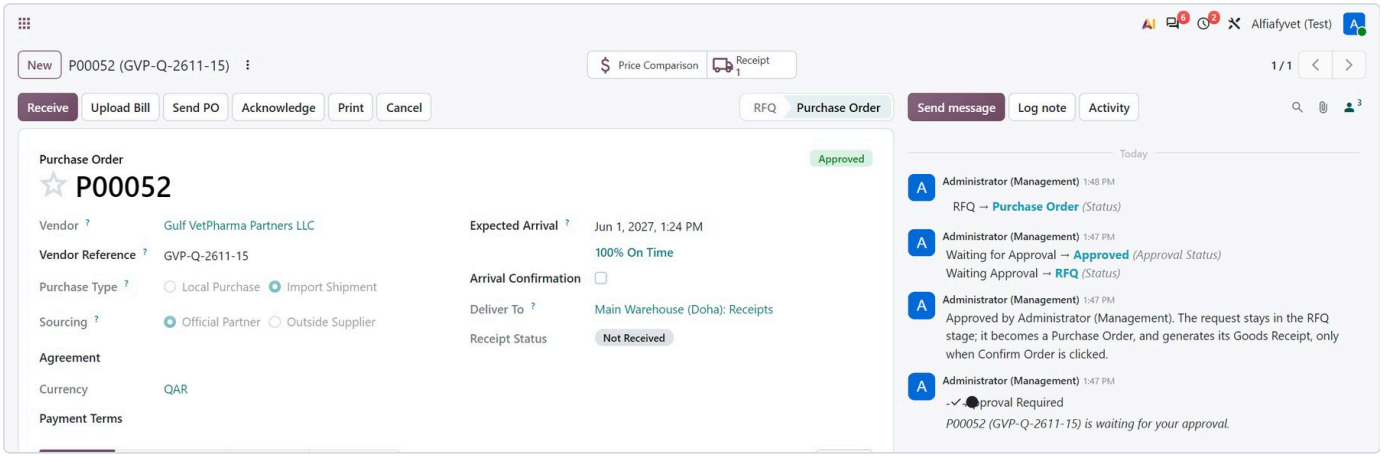
buyer (requester) on a PO waiting approval: no Approve button; Cancel asks for a reason (withdrawal).



accountant (approver) on the same PO: Approve button available.



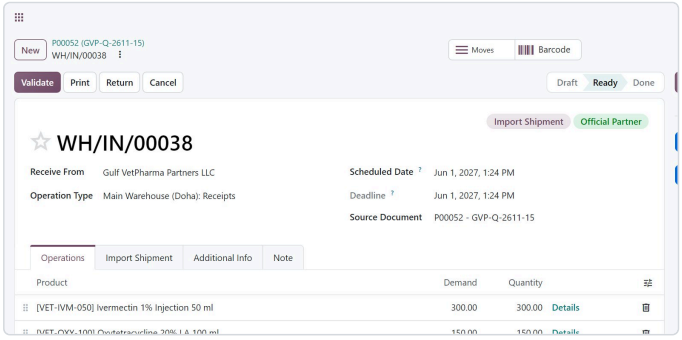
Approved: the RFQ returns to the RFQ stage with an "Approved" badge; the buyer now gets "Confirm Order".



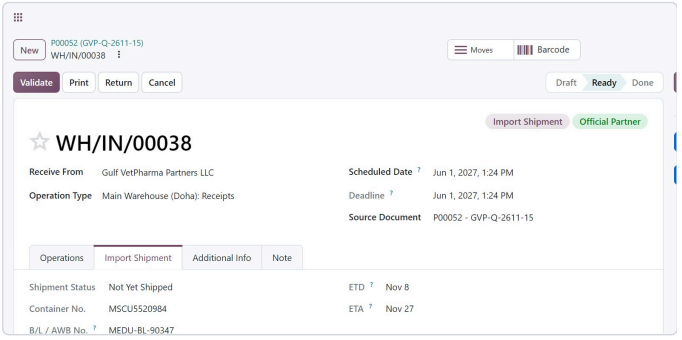
Confirmed purchase order with its receipt (smart button "Receipt").

6.5 Sourcing hand-over to receipts and lots

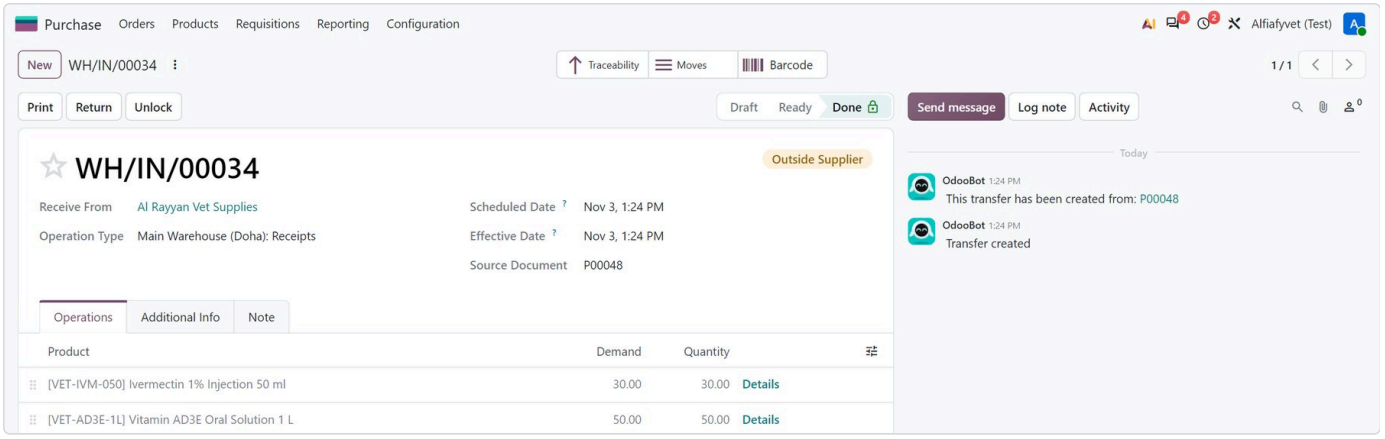
The receipt carries the PO sourcing (badge) and, for imports, the shipment data. When a vendor receipt is validated, every lot received is stamped Official Partner or Outside Supplier.



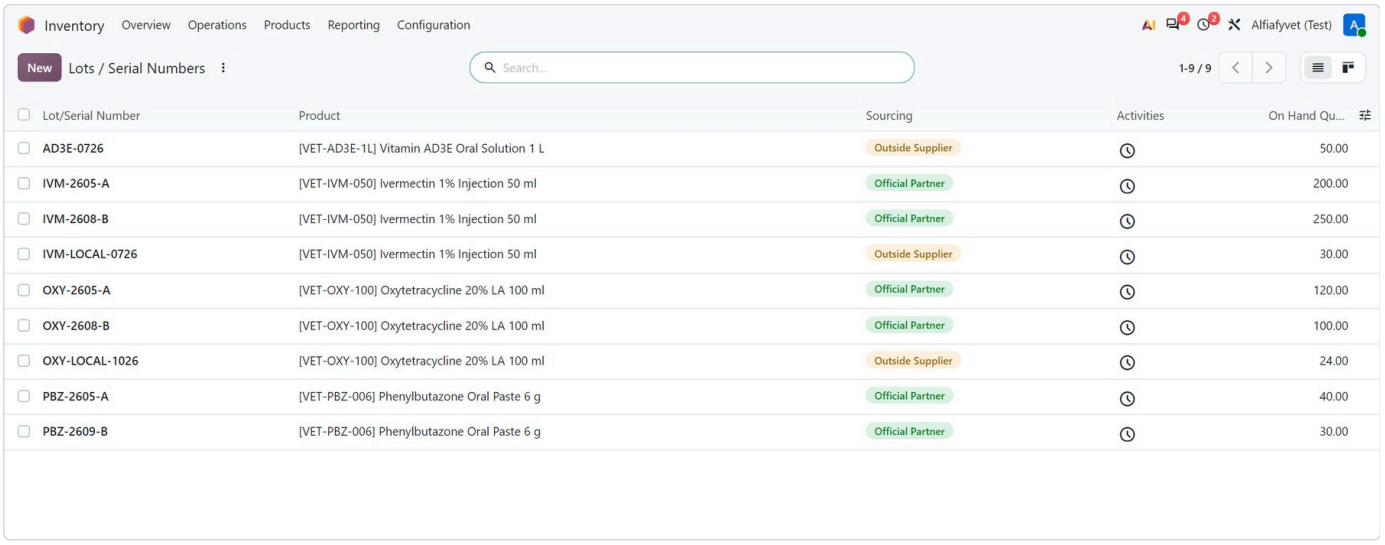
Receipt from the import PO: Import Shipment + Official Partner badges.



Import Shipment tab on the receipt (read from the PO).



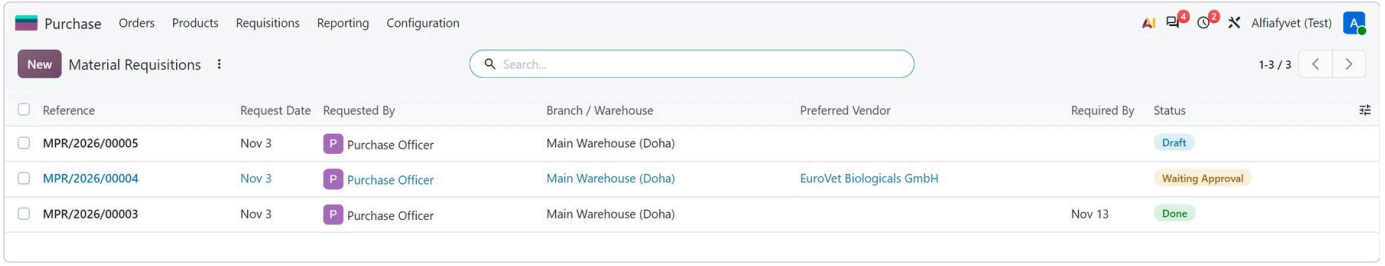
Validated receipt from the outside supplier (WH/IN/00034) with the Outside Supplier badge.



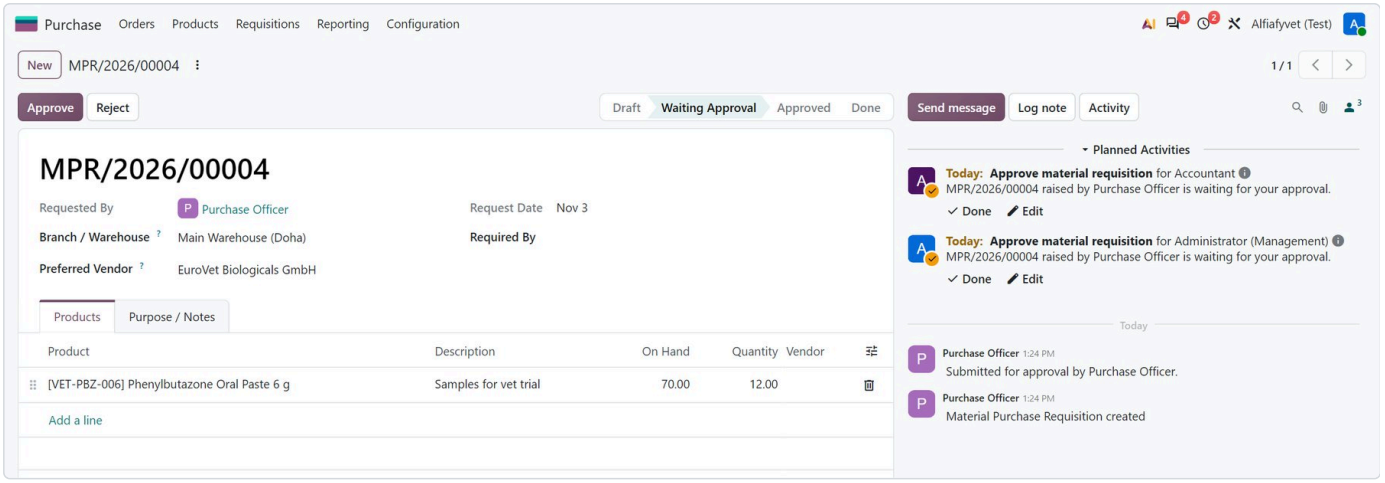
Lots / Serial Numbers: the same product (e.g. Ivermectin) has Official Partner lots (IVM-2605-A, IVM-2608-B) and an Outside Supplier lot (IVM-LOCAL-0726).

6.6 Material Purchase Requisitions

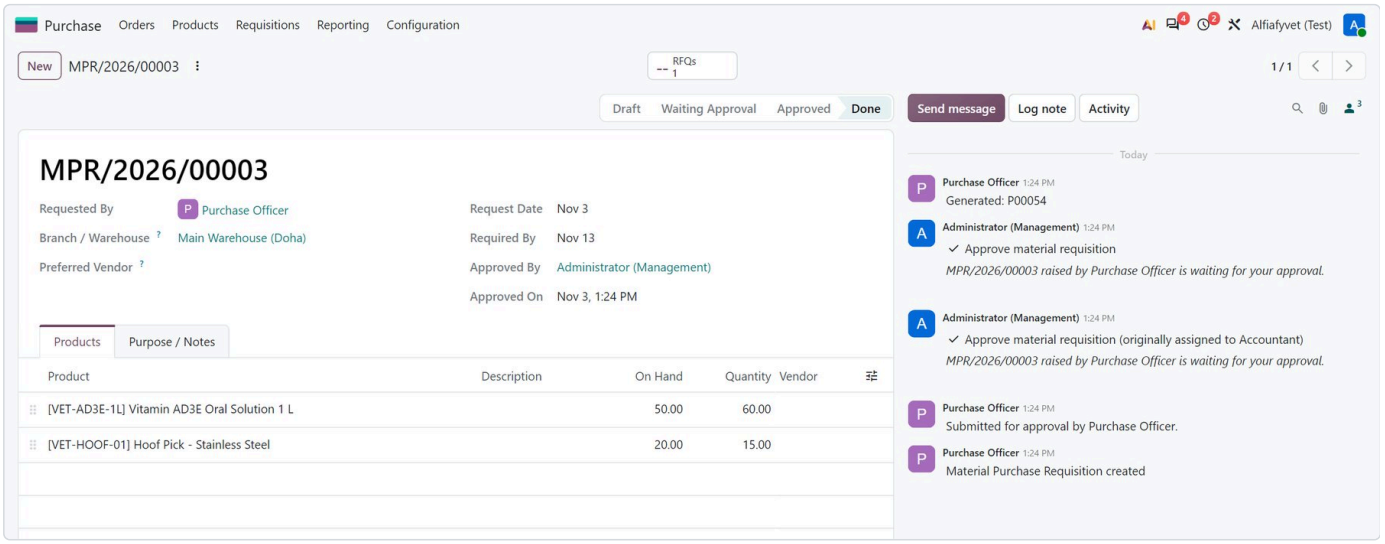
Purchase > Requisitions > Material Requisitions. Numbering MPR/YYYY/NNNNN. Draft → Waiting Approval (Purchase Administrators get an activity) → Approved → Done. “Create Local RFQ” creates one draft RFQ per vendor, tagged Local Purchase, receiving into the requisition’s warehouse. The RFQ then goes through the normal PO approval.



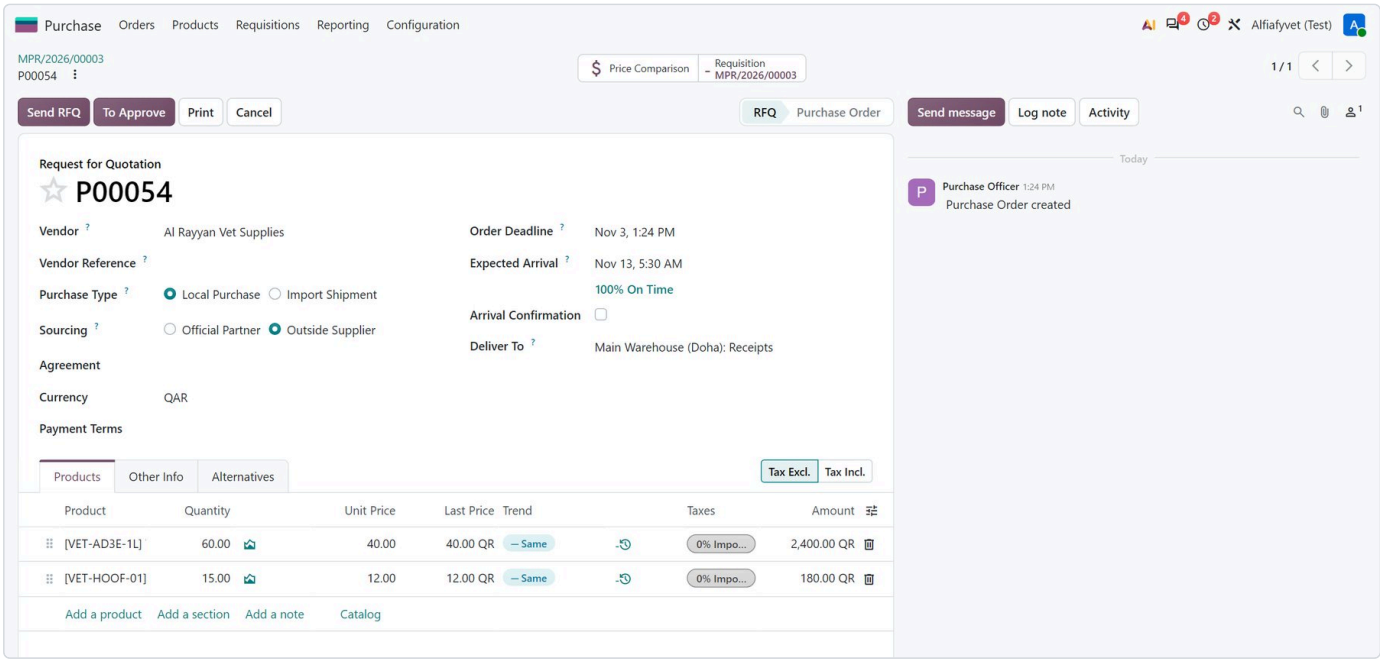
Requisition list in the three main states.



Requisition waiting approval: Approve / Reject for Purchase Administrators, activities for the approvers.

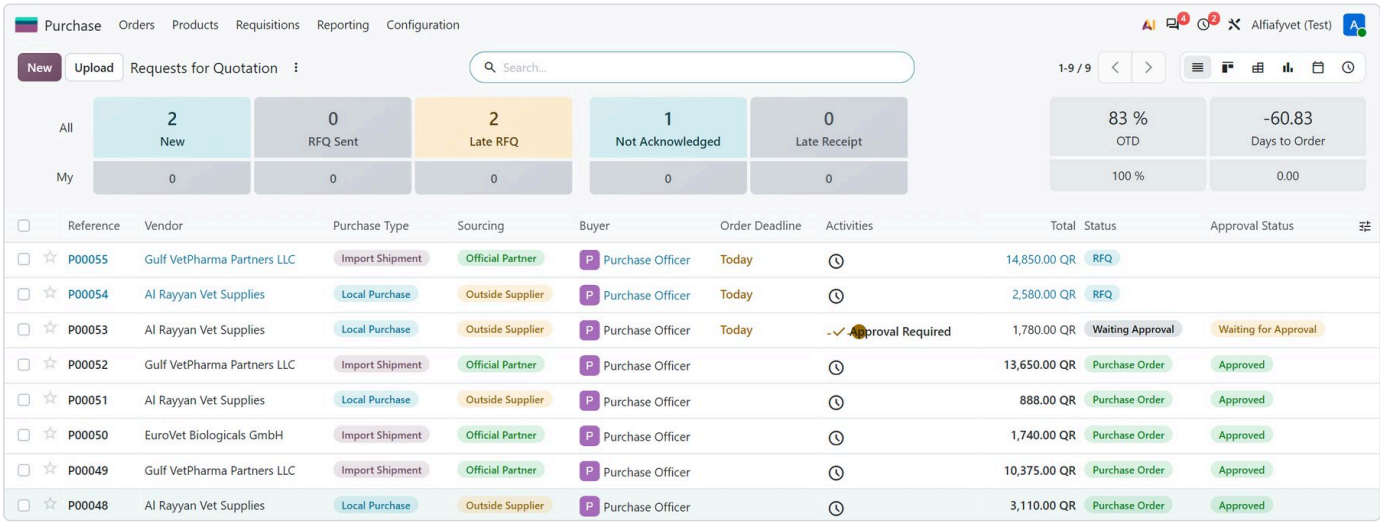


Requisition done: approved by / on, RFQ smart button, chatter trail.



Generated RFQ P00054: Local Purchase, vendor taken from the product Supplier (Al Rayyan, Outside Supplier), back-link to MPR/2026/00003, "To Approve" required.

6.7 Lists and filters



Requests for Quotation list: Purchase Type and Sourcing badges next to Approval Status. Filters “Local Purchases / Import Shipments / Official Partner / Outside Supplier” and group-by on type, sourcing and shipment status were added to the search.

7. Verification results

The seven checks of the implementation plan (section 3), each run as an automated test and repeated in the browser:

#	Check (implementation plan)	Automated test	Browser evidence	Result
1	Vendor auto-fill of purchase type and sourcing	test_01_vendor_autofill	Fig. s08, s09	**Pass**
2	Previous price loads; trend ▲ / ▼ / —	test_02_last_price_and_trend	Fig. s09	**Pass**
3	History popup: dates, vendors, prices	test_03_price_history_popup	Fig. s11	**Pass**
4	Import Tracking tab shows / hides	test_04_import_tracking_tab	Fig. s10 (import), s24 (local: no tab)	**Pass**
5	Receipt and lot carry Partner / Outside	test_05_sourcing_handover_to_receipt_and_lot	Fig. s15, s18, s19	**Pass**
6	Requisition: submit, approve, draft Local PO	test_06_material_requisition_flow	Fig. s20–s24	**Pass**
7	Unauthorised users cannot confirm without approval	test_07_approval_lock, test_08_sourcing_required_before_approval	Fig. s12, s26, s27	**Pass**

Full run: `-u restricted_cost_access,sale_and_purchase_approvals_steps,alfiafy_product_base,alfiafy_purchase --test-enable` → **0 failed, 0 error(s) of 22 tests**.

Module	Test	Result
alfiafy_product_base	`test_01_vet_fields_stored_on_template`	**Pass**
alfiafy_product_base	`test_02_search_by_molecule_arabic_name_and_species`	**Pass**
alfiafy_product_base	`test_03_stock_grouped_by_manufacturer_and_supplier`	**Pass**
alfiafy_product_base	`test_04_cost_hidden_from_branch_user`	**Pass**
alfiafy_product_base	`test_05_cost_visible_to_cost_manager`	**Pass**
alfiafy_product_base	`test_06_product_form_loads_for_both_users`	**Pass**
alfiafy_purchase	`test_01_vendor_autofill`	**Pass**
alfiafy_purchase	`test_02_last_price_and_trend`	**Pass**
alfiafy_purchase	`test_03_price_history_popup`	**Pass**
alfiafy_purchase	`test_04_import_tracking_tab`	**Pass**
alfiafy_purchase	`test_05_sourcing_handover_to_receipt_and_lot`	**Pass**
alfiafy_purchase	`test_06_material_requisition_flow`	**Pass**
alfiafy_purchase	`test_07_approval_lock`	**Pass**
alfiafy_purchase	`test_08_sourcing_required_before_approval`	**Pass**
restricted_cost_access	`test_grouped_aggregates_redacted`	**Pass**
restricted_cost_access	`test_order_margin_and_schema`	**Pass**
restricted_cost_access	`test_sale_form_hides_cost_columns`	**Pass**
restricted_cost_access	`test_sale_line_cost_and_margin`	**Pass**
restricted_cost_access	`test_stock_valuation_redacted`	**Pass**
sale_and_purchase_approvals_steps	`test_purchase_rejection_needs_reason`	**Pass**
sale_and_purchase_approvals_steps	`test_sale_order_needs_approval_above_threshold`	**Pass**
sale_and_purchase_approvals_steps	`test_small_sale_order_confirms_directly`	**Pass**

8. Notes and open points

- **Product fields to confirm**: the pasted DES00134 table stopped at `target_species`; see section 3.
- **Approvers**: the install creates the "all POs" step for the administrator. Replace / add the real management user and the accountant at sequence 10 (Settings > Technical > Approval Steps, developer mode).
- **Sourcing is mandatory** before "To Approve" and "Confirm Order", so lots are always classified. RFQs created by reordering rules for an unclassified vendor stop there until it is set: classify every vendor during data migration.
- **Material requisition**: the BRD leaves open who raises and approves requisitions. Currently purchase users raise them and Purchase Administrators approve; branch staff without Purchase access cannot raise one yet.
- **Last Price** on the PO line compares with the same vendor (as in the plan); the History popup shows all vendors.
- **Lot sourcing** is stamped once, at the first receipt of that lot; a later receipt of the same lot number from another source keeps the first value.
- The PO link to the requisition is `material_requisition_id`, because `requisition_id` is already used by Odoo's Purchase Agreements.
- Qatar has no VAT: the test company uses the generic chart, so products show a default 15% tax; this is only test data.
- Nothing was committed or pushed: the `testalfiafyvet` repository has no commits yet. The original 19.0 versions of the two upgraded modules were replaced in place.

How to run

Start the test server:

```
.venv20\Scripts\python.exe odoo\odoo-bin -c odoo\odoo.conf --addons-path=enterprise-20.0,odoo\addons,testalfiafyvet -d alfiafy20_test -p 8021
```

Re-run the tests:

```
... -d alfiafy20_test -u alfiafy_product_base,alfiafy_purchase,restricted_cost_access,sale_and_purchase_approvals_steps --test-enable --test-tags /alfiafy_product_base,/alfiafy_purchase,/restricted_cost_access,/sale_and_purchase_approvals_steps --stop-after-init
```

Install on another database: add `testalfiafyvet` to the addons path and install `alfiafy_purchase` (it pulls in the other three).